Notes on lecture 4

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1 Parametrization of a curve with respect to arc length

1.1 Re-parametrization of a curve

In the previous lecture, we have already seen the parametrization of a curve in terms of the parameter t. But this parametrization is not unique. Instead of t, we can use some other variable to parametrize the same curve. Eventually, any parametrization (that we use) of the curve (if it is accurate), will give us the same length of the curve.

Example 1.1.1. Suppose C is a curve which is parametrized by $\vec{r}(t) =< t, t^2, t^3 >$, where $1 \leq t \leq 2$. Suppose we define $t = e^u$. Then we can **re-parametrize** the same curve C as $\vec{r}(u) =< e^u, e^{2u}, e^{3u} >$, where $0 \leq u \leq ln\ 2$. □

1.2 The arc length function

Let us consider a curve $\vec{r}(t) =< x(t), y(t), z(t) >$ in space, where $t \in [a, b]$. So the initial point of the curve is $\vec{r}(a)$. Earlier, we were considering the length of the whole curve. Suppose now, we would like to consider only a part of this curve, that is, an arc (from the initial point to some other point on the curve). Suppose we take a variable point $t \in [a, b]$ and we simply want to find out the length of the curve traced out from a to t.
Earlier, the length of the whole curve was given by $L = \int_a^b |\vec{r'}(t)|dt$. But if we are considering only the part of the curve traced out from a to t, then we consider the integral only from a to t (NOT from a to b as considered for

getting the length of the whole curve), and we define

$$s(t) := \int_a^t |\vec{r'}(u)| du. \qquad (***)$$

This expression $s(t)$ is going to give us the length of the curve traced out from a to t. The expression on the right hand side of $(***)$ above is a function of t, that's why we have defined s to be a function of the variable t. This function $s(t)$ is known as the **arc length function**.

Remark 1.2.1. Observe that if we apply the fundamental theorem of integral calculus on $(***)$, then we get

$$\frac{ds}{dt} = |\vec{r'}(t)|.$$

Remark 1.2.2. Whenever we talk about arc length or re-parametrization of a curve (in books or in exercises), if nothing is stated, you should invariably assume that it is "in the increasing direction of t".

1.3 Re-parametrizing a curve with respect to the arc length function

In subsection 1.2 above, we saw that the arc length function s is a function of the variable t. Sometimes (not always), we may be able to express t explicitly in terms of s. In that case, we are going to obtain a way of re-parametrizing the same curve with respect to the arc length function s. Please see Example 1.3.1 below.

Example 1.3.1. Consider the curve $\vec{r}(t) =< \cos t, \sin t, t >$, where $t \in [0, 2\pi]$. Suppose we want to re-parametrize this curve with respect to the arc length function s.
Observe that $\vec{r'}(t) =< -\sin t, \cos t, 1 >$. This implies that $|\vec{r'}(t)| = \sqrt{2}$. Now $(***)$ tells us that

$$s(t) = \int_0^t \sqrt{2} du = \sqrt{2}t.$$

This implies that $t = \frac{s}{\sqrt{2}}$, where we denote $s(t)$ by s. So, in this example, we can express t explicitly in terms of s. Hence we get

$$\vec{r}(t) = \vec{r}(t(s)) =< \cos(\frac{s}{\sqrt{2}}), \sin(\frac{s}{\sqrt{2}}), \frac{s}{\sqrt{2}} >,$$

where $s \in [0, 2\sqrt{2}\pi]$. So we have re-parametrized the same curve with respect to s. $\square$

Homework: Re-parametrize the following curves with respect to arc length measured from the point where $t = 0$ in the direction of increasing t:

- $\vec{r}(t) = 2t\hat{i} + (1 - 3t)\hat{j} + (5 + 4t)\hat{k}$.
- $\vec{r}(t) = e^{2t}\cos 2t\hat{i} + 2\hat{j} + e^{2t}\sin 2t\hat{k}$.

2 Curvature

Consider a curve C in the 3 dimensional space. Let the curve be such that at each point lying on the curve, we can draw a unique tangent. Then the curve is called **smooth**. See Figure 2.0.1 for an illustration.

The prerequisites for the smoothness of a curve C (where $\vec{r}(t)$ is a parametriza-

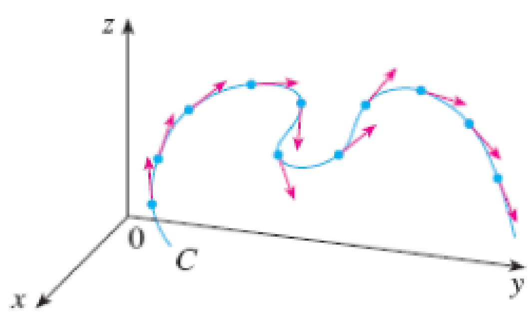


Figure 2.0.1: A smooth curve

tion of C) are:

- $\vec{r'}(t)$ exists and is continuous and
- $\vec{r'}(t) \neq \vec{0}$.

So if $\vec{r}(t)$ is a parametrization of a curve C, and if, the conditions

- $\vec{r'}(t)$ exists and is continuous and
- $\vec{r'}(t) \neq \vec{0}$

are satisfied, then we say that the **parametrization of the curve is smooth**. In order to be smooth, the curve should not have any sharp corners or cusps.
We will now discuss curvature. Whenever we talk about curvature, we will talk about it for a smooth curve only. The curvature is a measure of how quickly a curve changes its direction.
Suppose we have a smooth curve in space. Let $\vec{r}(t)$ be a parametrization of the curve. We know that the unit tangent to the curve is simply defined to be

$$\vec{T} = \frac{\vec{r'}(t)}{|\vec{r'}(t)|}.$$

The **curvature** (of the curve) is denoted by the symbol κ and is defined as

$$\kappa = \left| \frac{d\vec{T}}{ds} \right|,$$

where $s = s(t)$ denotes the arc length of the curve from the initial point upto the point corresponding to t. That is, the curvature κ of the curve is the magnitude of the rate of change of $\vec{T}$ with respect to the arc length s.
Observe that

$$\kappa = \left| \frac{d\vec{T}}{ds} \right| = \left| \frac{\frac{d\vec{T}}{dt}}{\frac{ds}{dt}} \right|,$$

where $s = \int_a^t |\vec{r'}(u)| du$. We know from the previous lecture that $\frac{ds}{dt} = |\vec{r'}(t)|$. Hence

$$\kappa = \left| \frac{\frac{d\vec{T}}{dt}}{\frac{ds}{dt}} \right| = \left| \frac{\vec{T'}(t)}{\vec{r'}(t)} \right|. \qquad (*)$$

We are going to use the last expression $(*)$ (for κ) later on in order to derive some formulae for the curvature.

Example 2.0.1. Find the curvature of a circle of radius a.
Whenever we want to define $\vec{r}(t)$ for a circle of radius a lying on a plane, we are going to have:

$$\vec{r}(t) =< a\cos t, a\sin t > .$$

This implies that

$$\vec{r'}(t) =< -a\sin t, a\cos t > .$$

So

$$\vec{T}(t) = \frac{\vec{r'}(t)}{|\vec{r'}(t)|}$$

$$= \frac{< -a\sin t, a\cos t >}{a} =< -\sin t, \cos t > .$$

This implies that $\vec{T'}(t) =< -\cos t, -\sin t >$. Clearly $|\vec{T'}(t)| = 1$ and $|\vec{r'}(t)| = a$. So for a circle of radius a, we have:

$$\kappa = \left|\frac{\vec{T'}(t)}{\vec{r'}(t)}\right| = \frac{1}{a}.$$

□

Remark 2.0.2. In Example 2.0.1 above, we saw that the curvature of a circle of radius a equals $\frac{1}{a}$. That means larger the radius of the circle, lesser the curvature. Suppose there are two circular roads of radii a_1 and a_2, where $a_2 >> a_1$. Now if we consider the curvature at some point P_1 (on the circle of radius a_1) and at some point P_2 (on the circle of radius a_2), then we can see that on the outer circle, the curvature is less.

Theorem 2.0.3. *The curvature of a curve given by the vector function $\vec{r}(t)$ is*

$$\kappa(t) = \frac{|\vec{r'}(t) \times \vec{r''}(t)|}{|\vec{r'}(t)|^3}.$$

PROOF: We know that $\vec{T}(t) = \frac{\vec{r'}(t)}{|\vec{r'}(t)|}$. We also know that $\frac{ds}{dt} = |\vec{r'}(t)|$, where s denotes the arc length function. This implies that

$$\vec{T}(t) = \frac{\vec{r'}(t)}{ds/dt}.$$

In other words,

$$\vec{r'}(t) = \frac{ds}{dt}\vec{T}(t).$$

$$\Rightarrow \vec{r''}(t) = \frac{d^2s}{dt^2}\vec{T} + \frac{ds}{dt}\vec{T'}.$$

Now, from the properties of vector cross product, we know that if $\vec{a}.\vec{b}, \vec{c}$ are 3 vectors and $\lambda_1, \lambda_2, \lambda_3$ are scalars, then

$$\lambda_1\vec{a} \times (\lambda_2\vec{b} + \lambda_3\vec{c}) = (\lambda_1\lambda_2)\vec{a} \times \vec{b} + (\lambda_1\lambda_3)\vec{a} \times \vec{c}.$$

So

$$\vec{r'}(t) \times \vec{r''}(t) = \left(\frac{ds}{dt}\right)\left(\frac{d^2s}{dt^2}\right)(\vec{T} \times \vec{T}) + \left(\frac{ds}{dt}\right)^2 (\vec{T} \times \vec{T'}) \qquad (**)$$

Now since the angle between $\vec{T}$ and itself is zero, it follows that $\vec{T} \times \vec{T} = \vec{0}$. We also know from one of the earlier lectures that if $\vec{u}$ is a vector of constant length, then $\vec{u}$ and $\vec{u'}$ are orthogonal. Since $\vec{T}$ is a unit vector, it is of constant length 1. Hence $\vec{T}$ and $\vec{T'}$ are orthogonal. So we get from $(**)$ that:

$$\begin{aligned} |\vec{r'}(t) \times \vec{r''}(t)| &= \left(\frac{ds}{dt}\right)^2 |\vec{T}||\vec{T'}| \sin\frac{\pi}{2} \\ &= \left(\frac{ds}{dt}\right)^2 |\vec{T'}|, \end{aligned}$$

as $|\vec{T}| = 1$. Now since $\frac{ds}{dt} = |\vec{r'}(t)|$, we get from the above that

$$|\vec{T'}| = \frac{|\vec{r'}(t) \times \vec{r''}(t)|}{|\vec{r'}(t)|^2}.$$

So

$$\kappa(t) = \frac{|\vec{T'}|}{|\vec{r'}(t)|} = \frac{|\vec{r'}(t) \times \vec{r''}(t)|}{|\vec{r'}(t)|^3}.$$

□

Theorem 2.0.4. *For a curve $y = f(x)$ in the cartesian coordinate plane,*

$$\kappa(x) = \frac{|f''(x)|}{[1 + \{f'(x)\}^2]^{3/2}}.$$

PROOF: Let $\vec{r}(t)$ denote the position vector of any point $(x(t), y(t))$ on the curve. Since we are in the cartesian coordinate system, we can write

$$\vec{r} =< x, y > .$$

Since the curve is given by $y = f(x)$, we can further write:

$$\vec{r} =< x, f(x) > .$$

This implies that

$$\vec{r'} =< 1, f'(x) > .$$

Now, if we simply follow the steps in the proof of Theorem 2.0.3 above, we will get the proof of this theorem. This is left to you as an **exercise**. □

3 Normal and Binormal vectors

Suppose we are considering a curve C in the 3 dimensional space. Let C be parametrized by $\vec{r}(t) =< x(t), y(t), z(t) >$. Let P denote the point $(x(t), y(t), z(t))$ on the curve C. Then $\vec{r'}(t)$ is the tangent to the curve C at the point P. Now there are infinitely many vectors which are perpendicular to the tangent vector $\vec{r'}(t)$ in various directions. The unit tangent vector in the direction of $\vec{r'}(t)$ is given by $\vec{T} = \frac{\vec{r'}(t)}{|\vec{r'}(t)|}$. Throughout this section, we will assume that $\vec{r'}(t) \neq \vec{0}$ (for the tangent vector at the point P to exist).
Now since $\vec{T}$ is a unit vector, it has constant length 1. So we know from one of the previous lectures that $\vec{T}$ and $\vec{T'}$ will be orthogonal to each other. This means that we will always be able to find out a unique direction (namely, the direction of the vector $\vec{T'}$), which is perpendicular to the tangent vector $\vec{r'}(t)$. So, we are going to choose the direction of $\vec{T'}$ as **the direction of the normal vector**, which is perpendicular to the unit tangent vector $\vec{T}$.

Remark 3.0.1. The vector $\vec{T}$ is a unit vector does not imply that the vector $\vec{T'}$ is also a unit vector.

Take
$$\vec{N} = \frac{\vec{T'}}{|\vec{T'}|}.$$
Then $\vec{N}$ is a unit vector. Also, $\vec{T}$ and $\vec{N}$ are perpendicular to each other. This vector $\vec{N}$ is called the **principal unit normal vector** to the curve C at the point P.
The vector $\vec{T} \times \vec{N}$ is perpendicular to both $\vec{T}$ and $\vec{N}$. Let
$$\vec{B} = \vec{T} \times \vec{N}.$$
So the vectors $\vec{T}, \vec{N}$ and $\vec{B}$ form a coordinate system at the point P (you can consider a local coordinate system by considering 3 axes along the 3 directions $\vec{T}$, $\vec{N}$, and $\vec{B}$), and this vector $\vec{B}$ is called the **binormal vector** to the curve C at the point P. Figure 3.0.1 illustrates this.

Example 3.0.2. Let $\vec{r}(t) =< \cos t, \sin t, t >$. Then $\vec{r}$ denotes a helix. And
$$\vec{T} = \frac{\vec{r'}(t)}{|\vec{r'}(t)|} = \frac{< -\sin t, \cos t, 1 >}{\sqrt{2}}.$$

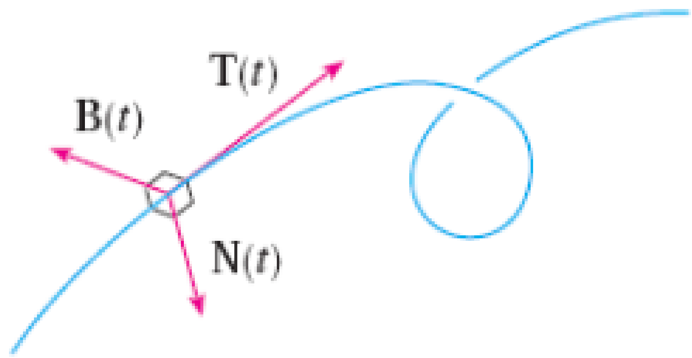


Figure 3.0.1: The normal and binormal vectors

Hence

$$\vec{T'} = \frac{1}{\sqrt{2}} < -\cos t, -\sin t, 0 > \text{ and } |\vec{T'}| = \frac{1}{\sqrt{2}}.$$

The unit normal vector $\vec{N}$ is then given by

$$\vec{N} = \frac{\vec{T'}}{|\vec{T'}|} =< -\cos t, -\sin t, 0 > .$$

Also $\vec{B} = \vec{T} \times \vec{N} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -\frac{\sin t}{\sqrt{2}} & \frac{\cos t}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\cos t & -\sin t & 0 \end{vmatrix}$. □

4 Motion in space: Velocity and acceleration

Consider the space curve C in Figure 4.0.1. Let $\vec{r}(t)$ be a parametrization of C. If we consider the curve C to be the trajectory of a particle moving in space, and if the parameter t denotes time, then the derivative

$$\frac{d\vec{r}}{dt} = \lim_{h \to 0} \frac{\vec{r}(t+h) - \vec{r}(t)}{h}$$

is nothing but the **velocity** vector $\vec{v}$ of the particle. In Figure 4.0.1, the point P denotes the position of the particle at time t, and the point Q denotes the position of the particle after time h. Let $\vec{r}(t) =< x(t), y(t), z(t) >$. Then

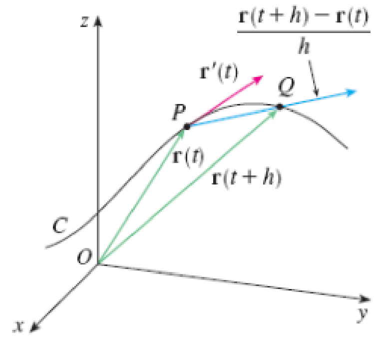


Figure 4.0.1: Motion of a particle in space

from one of the earlier lectures, we know that $\frac{d\vec{r}}{dt} =< \frac{dx}{dt}, \frac{dy}{dt}, \frac{dz}{dt} >$. So

$$\vec{v} =< \frac{dx}{dt}, \frac{dy}{dt}, \frac{dz}{dt} > .$$

Again, the magnitude $|\vec{v}|$ of $\vec{v}$ is nothing but the **speed** of the particle. Since, we already know from one of the previous lectures that $|\vec{r'}(t)| = \frac{ds}{dt}$ (where s denotes the arc length function), it follows that

$$|\vec{v}| = |\vec{r'}(t)| = \frac{ds}{dt}.$$

This simply means that if we consider the arc length function s, then $\frac{ds}{dt}$ is nothing but the speed. Also, the **acceleration** is given by

$$\frac{d\vec{v}}{dt} = \frac{d^2\vec{r}}{dt^2} =< \frac{d^2x}{dt^2}, \frac{d^2y}{dt^2}, \frac{d^2z}{dt^2} > .$$

5 Functions of several variables

If we want to express some physical phenomenon mathematically, then it always involves at least two variables, namely, space and time. We will now study functions of more than one variables. Let us start with functions of 2 variables.

5.1 Functions of two variables

Let x and y be the two variables. Let $\mathbb{R}^2$ denote the set of all ordered pairs (x, y), where x, y belong to $\mathbb{R}$ (Here $\mathbb{R}$ denotes the real line.). Let D be a subset of $\mathbb{R}^2$.

Definition 5.1.1. A **function** $f : D \to \mathbb{R}$ is a rule which assigns to each point $(x, y) \in D$, a unique point $f(x, y)$ in $\mathbb{R}$. The set D is called the **domain** of the function f, and the subset of $\mathbb{R}$ in which the points $f(x, y)$ belong for all $(x, y) \in D$ is called the **range** of f. We denote the range of f by the symbol R_f. So R_f is given by

$$R_f = \{z \in \mathbb{R} | z = f(x, y) \text{ where } (x, y) \in D \subseteq \mathbb{R}^2\}.$$

□

Remark 5.1.2. We often write $z = f(x, y)$ to make explicit the value taken on by f at the general point (x, y). In Definition 5.1.1 above, x and y are independent variables, and z is the dependent variable. The domain D of f is a subset of $\mathbb{R}^2$ and the range R_f of f is a subset of $\mathbb{R}$. See Figure 5.1.1 for an illustration.

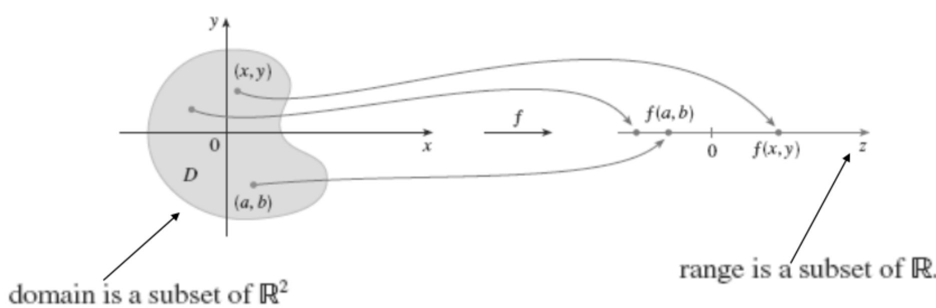


Figure 5.1.1: Functions of two variables

Example 5.1.3. For each of the following functions, evaluate $f(3,2)$ and sketch the domain of f:
(a) $f(x,y) = \frac{\sqrt{x+y+1}}{x-1}$.
(b) $f(x,y) = x\ ln(y^2 - x)$.
To find out the domain of f, we have to find out all $(x,y) \in \mathbb{R}^2$ where the function f is well defined.
For the function in (a), we can easily that f is well defined only when $x+y+1 \geq 0$ and $x-1 \neq 0$. So the domain of f in this case is the set of all points in $\mathbb{R}^2$ lying towards the right hand side of the straight line $x+y+1=0$ (including this line), and excluding the points on the straight line $x=1$. Figure 5.1.2 depicts this.
To find the domain of the function in part (b), observe that in this case, f

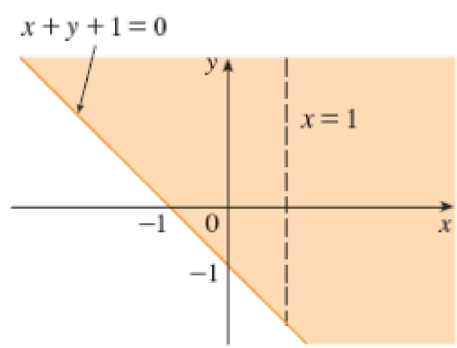


Figure 5.1.2: Domain of f in Example 5.1.3(a)

is well defined only when $y^2 - x \geq 0$, that is, when $x \leq y^2$. So the domain of f is shown by the shaded region in Figure 5.1.3.
One can easily calculate $f(3,2)$ for the functions given in (a) and (b) both.
□

Example 5.1.4. Find the domain and range of h where $h(x,y) = \sqrt{9 - x^2 - y^2}$.
For the function h to be well defined, we must have $9 - x^2 - y^2 \geq 0$, that is, we must have $x^2 + y^2 \leq 9$. So the domain D of h is given by:

$$D = \{(x,y) \in \mathbb{R}^2 | x^2 + y^2 \leq 9\}.$$

The range R_h of h is by definition the set $\{z \in \mathbb{R} | z = h(x,y), (x,y) \in D\}$. Observe that if $(x,y) \in D$, then $9 - x^2 - y^2 \leq 9$. This implies that if $(x,y) \in$

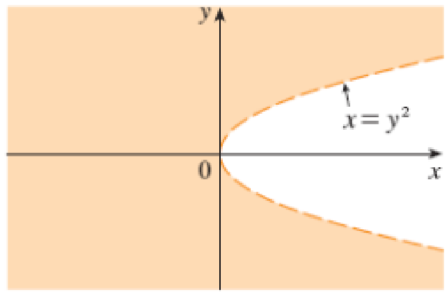


Figure 5.1.3: Domain of f in Example 5.1.3(b)

D, then $z = \sqrt{9 - x^2 - y^2} \leq 3$ (We are considering the positive square root here. Throughout this course, whenever we consider square roots, they will be positive square roots only, unless specifically mentioned). Clearly, since we are considering the positive square root, we must also have $z \geq 0$. So we can now see that for any $(x, y) \in D$, $0 \leq z = h(x, y) \leq 3$. Hence $R_h = [0, 3]$.
$\square$

5.2 Graph of a function of two variables

Consider a function f of a single variable x over an interval $[a, b]$. Then the **graph** G_f of f is given by:

$$G_f = \{(x, y) \in \mathbb{R}^2 | y = f(x), x \in [a, b]\}.$$

Clearly, when f is a function of a single variable x, $G_f \subseteq \mathbb{R}^2$.
Now consider a function f of two variables x and y. Let

$$f : D \to \mathbb{R}$$

be a function of two variables, where $D \subseteq \mathbb{R}^2$ is the domain of f. Then the **graph** G_f of f is given by:

$$G_f = \{(x, y, z) \in \mathbb{R}^3 | z = f(x, y), (x, y) \in D \subseteq \mathbb{R}^2\}.$$

Clearly in this case, we have $G_f \subseteq \mathbb{R}^3$. See Figure 5.2.1 for an illustration. In Figure 5.2.1, we have done the following:

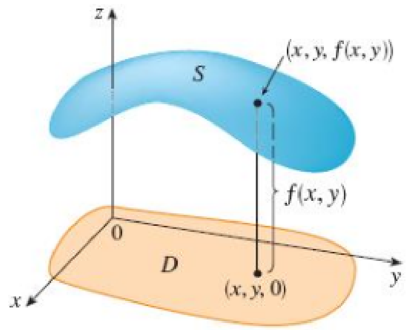


Figure 5.2.1: Graph of a function of two variables

For each point (x, y) in the domain D of f (that is, for each point $(x, y, 0)$ in the xy-plane, where $(x, y) \in D$), we have plotted a point in the space at a height $z = f(x, y)$ above it.

Now if we consider all the points of the form (x, y, z) where $z = f(x, y)$ and $(x, y) \in D$, we will obtain a **surface** S in the space (as shown in Figure 5.2.1). This surface S is the **graph** of the function f of two variables.

Remark 5.2.1. In Figure 5.2.1, everything has been drawn in the upper half of the xyz-space. But it is not always necessary that the figure has to lie in the upper half.

Example 5.2.2. Sketch the graph of the function $f(x, y) = 6 - 2x - 3y$.
Let $z = f(x, y) = 6 - 2x - 3y$. Then we have:

$$2x + 3y + z = 6.$$

If we rewrite the above equation in the form $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$, then it will represent a plane in space which intersects the x-axis at $a = 3$ units, the y-axis at $b = 2$ units, and the z-axis at $c = 6$ units. This is because $2x + 3y + z = 6$ is nothing but $\frac{x}{3} + \frac{y}{2} + \frac{z}{6} = 1$. Figure 5.2.2 depicts this. □

Example 5.2.3. Sketch the graph of the function $g(x, y) = \sqrt{9 - x^2 - y^2}$.
Let $z = g(x, y) = \sqrt{9 - x^2 - y^2}$. Then we have:

$$x^2 + y^2 + z^2 = 9.$$

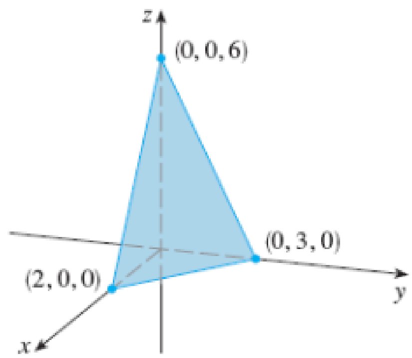


Figure 5.2.2: Graph of the function in Example 5.2.2

This is the equation of a sphere centered at the origin with radius 3.
Now, if we consider any point (x, y) in the xy-plane lying on the disc of radius 3 centered at the origin, then corresponding to that point (x, y), we will get two values of z on the sphere (Observe that if we draw a vertical line at the point (x, y) which is parallel to the z-axis, then that line will intersect the sphere at 2 points.). This violates the definition of a function, because corresponding to each point in the domain of definition of a function, we should get a unique point in the range. That's why we have to consider only the positive square root of $9 - x^2 - y^2$ while considering the function $g(x, y)$. This will give us the upper hemisphere as the graph of the function g. Figure 5.2.3 depicts this. Note that if the function had been $g(x, y) =$

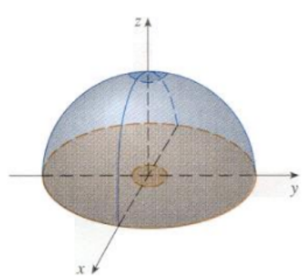


Figure 5.2.3: Graph of the function in Example 5.2.3

$-\sqrt{9 - x^2 - y^2}$, the lower hemisphere would have been the graph of g. $\square$

So we have now understood that the graph of a function of 2 variables can be visualized as a surface. Figure 5.2.4 shows some functions of 2 variables and their graphs.

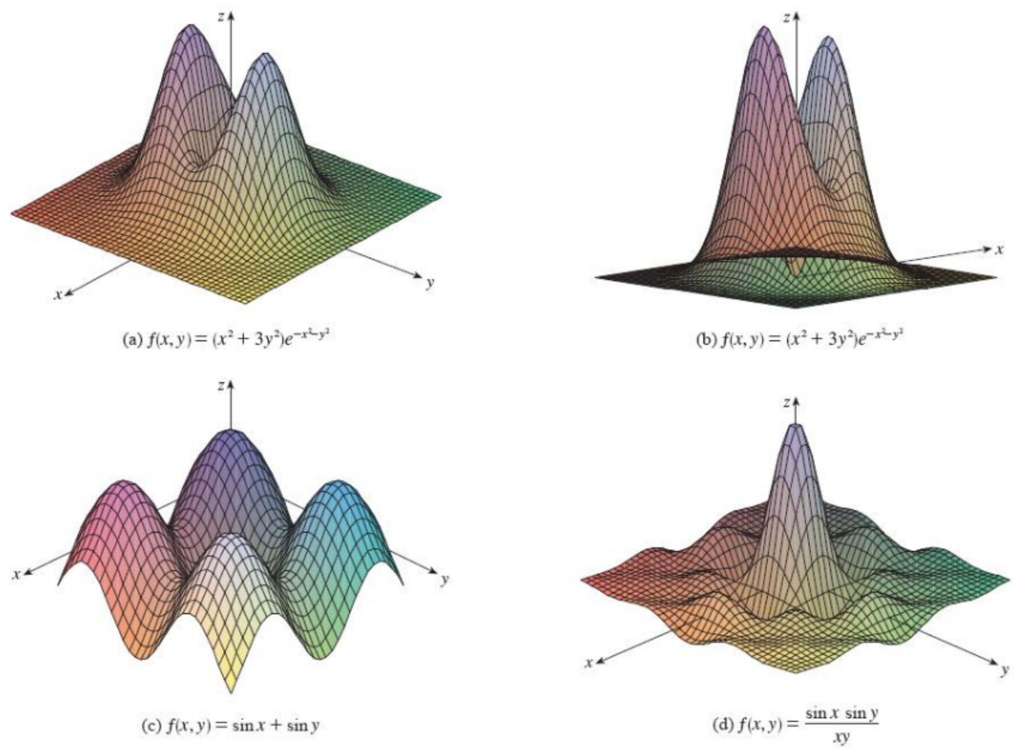


Figure 5.2.4: Graphs of some functions

Remark 5.2.4. Whenever the given function of 2 variables has $e^{-x^2-y^2}$ or $e^{x^2+y^2}$ in it, we are going to get a "bell-shaped" surface as a graph. See (a) and (b) of Figure 5.2.4 for an illustration.

5.3 Contour curves and Level curves

Motivation: we are considering the map of India in the xy-plane, and the temperature in various places of India is denoted by $z = T(x, y)$. Then we will get a surface in the space as the graph of the function $T(x, y)$. If we cut this surface by the $z = k$ plane (where k is a constant), then we will get a **curve** on the surface, which represents all places in India having the same temperature k. If we take the projection of such curves on the xy-plane, then we will get some curves in the xy-plane (that is, on the map of India) called **isotherms**. Isotherms are nothing but examples of level curves, which we will define soon.
Example: Let $z = x^2 + y^2$. Then this equation represents a paraboloid. If we cut the paraboloid by the $z = k$ plane (where k is a constant), then the plane cuts the paraboloid in the circle $x^2 + y^2 = k$ (Observe here that the plane $z = k$ is parallel to the xy-plane.). Take the projection of the circle $x^2 + y^2 = k$ on the xy-plane. Then we will get the circle on the xy-plane. If $k_1 > k$ and we do the same thing with the plane $z = k_1$, then we are going to get a bigger circle as a projection on the xy-plane. Such circles (which are the projections) on the xy-plane are examples of level curves.

Definition 5.3.1. Let $f(x, y)$ be a function of 2 variables. Let k be a constant in the range of f. Then the curve obtained by considering the intersection of the plane $z = k$ with the graph of $f(x, y)$ is called a **contour curve**. And the projections of the contour curves on the xy-plane are called **level curves**. $\square$

Remark 5.3.2. Contour curves are curves along the surface formed by the graph of the given function of 2 variables. Level curves are projections of the contour curves on the xy-plane.

Remark 5.3.3. In Definition 5.3.1 above, it is important that the constant k must lie in the range of f.

Figure 5.3.1 illustrates the concepts of a contour curve and a level curve. Figure 5.3.2 depicts the contour curves and the level curves for a hill.

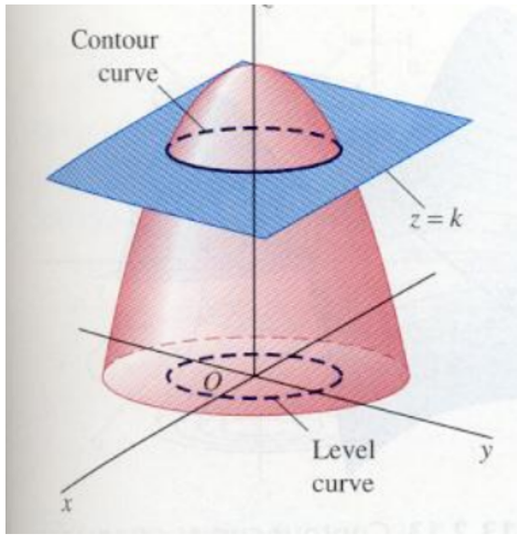


Figure 5.3.1: Contour curve and level curve

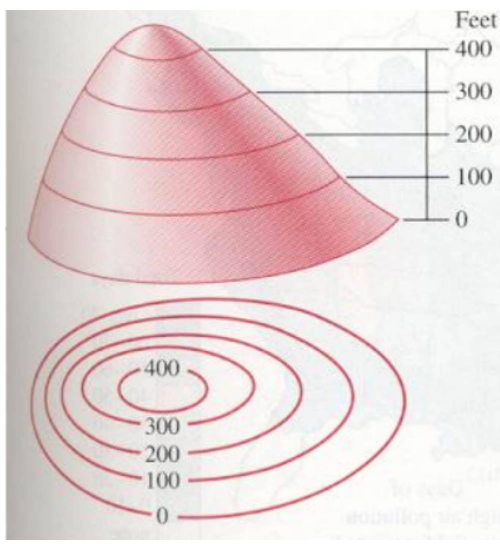
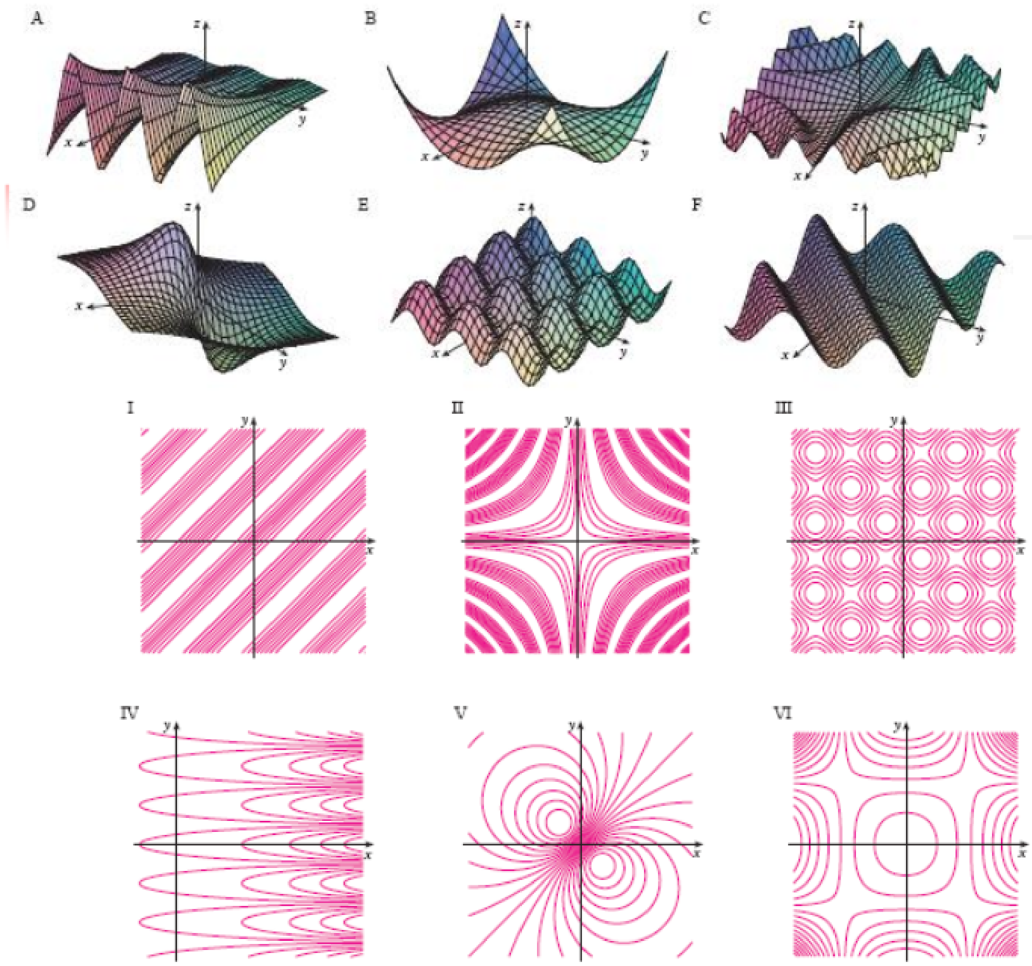


Figure 5.3.2: Contour curves and level curves for a hill

Figure 5.3.3 gives an **exercise** on level curves.



Can you Match the Surfaces And Their Level Curves?

7

Figure 5.3.3: An exercise

5.4 Functions of three or more variables

A function f of 3 variables, is a rule that assigns to each ordered triple (x, y, z) in a domain $D \subseteq \mathbb{R}^3$, a unique real number denoted by $f(x, y, z)$. For instance, the temperature T at any point on the surface of the earth depends on the longitude x, the latitude y of the point, and the time t. So we can write $T = f(x, y, t)$.
Suppose we have the function $f(x, y, z)$ (where $(x, y, z) \in D \subseteq \mathbb{R}^3$) and we want to find the graph of this function f. Observe that in this case, the

graph G_f of f is given by:

$$G_f = \{(x, y, z, u) \in \mathbb{R}^4 | u = f(x, y, z), (x, y, z) \in D \subseteq \mathbb{R}^3\}.$$

But it is not possible to visualize a graph in $\mathbb{R}^4$. However, if we take $f(x, y, z) = k$, where k is a constant belonging to the range of f, then it will represent a surface in $\mathbb{R}^3$. For example, if we take $x^2 + y^2 + z^2 = 9$, then it is going to represent a sphere of radius 3.
Just like in case of level curves, if for a function f of 3 variables x, y, z, we equate $f(x, y, z) = k$ (where k belongs to the range of f), then we are going to obtain a surface in $\mathbb{R}^3$. Such surfaces are called **level surfaces**. Figure 5.4.1 gives certain level surfaces for the function $f(x, y, z) = x^2 + y^2 + z^2$.
It is not possible to visualize the graph of a function f of 3 variables, since

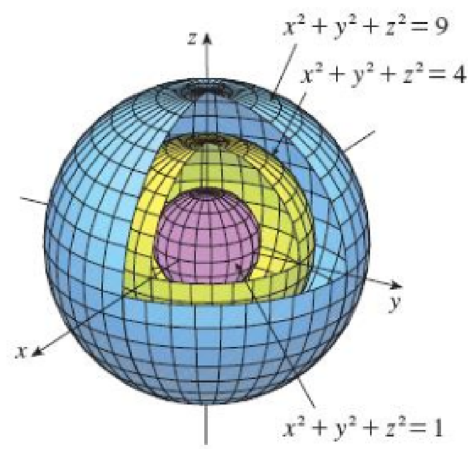


Figure 5.4.1: Level surfaces

that will lie in a 4 dimensional space. However, we do gain some insight about the function f by examining its level surfaces, which are the surfaces with equation $f(x, y, z) = k$, where k is a constant lying in the range of f. If the point (x, y, z) moves along a level surface, then the value of $f(x, y, z)$ remains fixed.

6 Limit of a function of two variables

Definition 6.0.1. Let f be a function of 2 variables whose domain $D \subseteq \mathbb{R}^2$ includes points arbitrarily close to the point (a, b). Then we say that the **limit** of $f(x, y)$ as (x, y) approaches (a, b) is L and we write

$$\lim_{(x,y)\to(a,b)} f(x, y) = L,$$

if for every number $\epsilon > 0$, there exists a number $\delta > 0$ such that

$$|f(x, y) - L| < \epsilon \text{ whenever } (x, y) \in D \text{ and } 0 < \sqrt{(x-a)^2 + (y-b)^2} < \delta.$$

□

6.1 Geometrical interpretation of limit

Let $D_\delta(a, b)$ denote the disc in the xy-plane having center at (a, b) and radius δ. Then the condition $0 < \sqrt{(x-a)^2 + (y-b)^2} < \delta$ simply means all those points in the domain D which are not equal to (a, b) and which lie in the interior of the disc $D_\delta(a, b)$. The definition of limit then says that for all $(x, y) \in D$ which are not equal to (a, b) and which lie in the interior of the disc $D_\delta(a, b)$, we must have that $f(x, y)$ lies in the interval $(L - \epsilon, L + \epsilon)$. Figure 6.1.1 explains this.

Figure 6.1.2 depicts the phenomenon in the 3 dimensional space. Observe

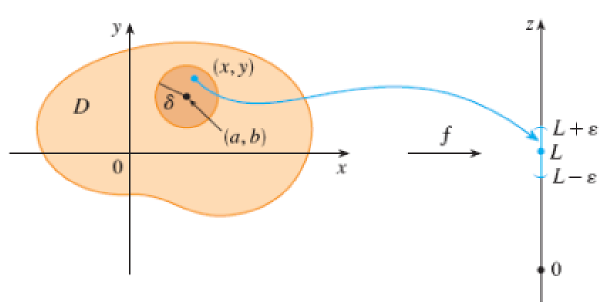


Figure 6.1.1: Geometrical interpretation of limit

that the disc on the surface S in Figure 6.1.2 is the image of $D_\delta(a, b)$ under the map f. The highest point and the lowest point on the surface S_δ

corresponding to the disc $D_\delta(a,b)$ (See Figure 6.1.2) must lie in the interval $(L-\epsilon, L+\epsilon)$.

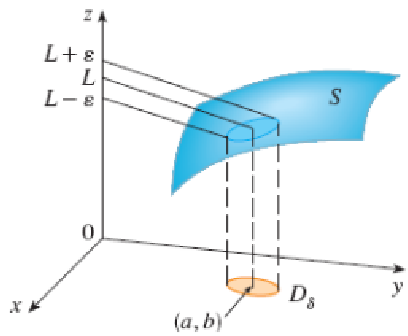


Figure 6.1.2: Limit of a function of 2 variables

6.2 Independence of path

In this subsection, we will see that the limit $\lim_{(x,y)\to(a,b)} f(x,y)$ exists if and only if whatever path we take in order to make the variable point (x,y) to approach the point (a,b), we obtain the same value of the limit. Let us begin with an example first.

Example 6.2.1. Let us compare the behavior of the functions $f(x,y)$ and $g(x,y)$ as (x,y) approaches the origin $(0,0)$, where

$$f(x,y) = \frac{\sin(x^2+y^2)}{x^2+y^2}$$

and

$$g(x,y) = \frac{x^2-y^2}{x^2+y^2}.$$

In Figure 6.2.1, we can see a table for some values of $f(x,y)$ as (x,y) approaches the origin $(0,0)$. Observe that in the table given in Figure 6.2.1, if one goes from -1 to 0 along the x and y axes both, then the value of $f(x,y)$ is approaching 1. The same is true if we go from 1 to 0 along the x and the y axes.

However, if we look at a similar table (Table 2 in Figure 6.2.2) for $g(x,y)$,

TABLE 1 Values of $f(x, y)$

$x \backslash y$	−1.0	−0.5	−0.2	0	0.2	0.5	1.0
−1.0	0.455	0.759	0.829	0.841	0.829	0.759	0.455
−0.5	0.759	0.959	0.986	0.990	0.986	0.959	0.759
−0.2	0.829	0.986	0.999	1.000	0.999	0.986	0.829
0	0.841	0.990	1.000		1.000	0.990	0.841
0.2	0.829	0.986	0.999	1.000	0.999	0.986	0.829
0.5	0.759	0.959	0.986	0.990	0.986	0.959	0.759
1.0	0.455	0.759	0.829	0.841	0.829	0.759	0.455

$$\lim_{(x,y)\to(0,0)} \frac{\sin(x^2+y^2)}{x^2+y^2} = 1$$

Figure 6.2.1: Table 1 for Example 6.2.1

we can see that as (x, y) approaches the origin, the values of $g(x, y)$ are not converging anywhere. So we can't conclude anything about the behavior of the function $g(x, y)$ near the origin.

So we can say intutively that

$$\lim_{(x,y)\to(0,0)} f(x, y) = 1$$

and

$$\lim_{(x,y)\to(0,0)} g(x, y) \text{ does not exist.}$$

□

Recall from single variable calculus that

$$\lim_{x\to a} f(x)$$

exists and equals L if and only if

$$\lim_{x\to a^+} f(x) = \lim_{x\to a^-} f(x) = L.$$

TABLE 2 Values of $g(x, y)$

$x \backslash y$	−1.0	−0.5	−0.2	0	0.2	0.5	1.0
−1.0	0.000	0.600	0.923	1.000	0.923	0.600	0.000
−0.5	−0.600	0.000	0.724	1.000	0.724	0.000	−0.600
−0.2	−0.923	−0.724	0.000	1.000	0.000	−0.724	−0.923
0	−1.000	−1.000	−1.000		−1.000	−1.000	−1.000
0.2	−0.923	−0.724	0.000	1.000	0.000	−0.724	−0.923
0.5	−0.600	0.000	0.724	1.000	0.724	0.000	−0.600
1.0	0.000	0.600	0.923	1.000	0.923	0.600	0.000

$$\lim_{(x,y)\to(0,0)} \frac{x^2 - y^2}{x^2 + y^2} \text{ does not exist}$$

Figure 6.2.2: Table 2 for Example 6.2.1

That is, it does not matter whether x approaches the point a from the left or the right hand side. In both the cases, the value of the function $f(x)$ approaches L.

In the case of functions of a single variable, the variable point x can approach the point a from the left and the right only. But for a function of 2 variables, this is not the case. There are infinitely many directions by which the variable point (x, y) can approach the point (a, b). Moreover, the variable point (x, y) can approach (a, b) not just through straight lines, but also via curves (A straight line can also be considered as a curve.).

Suppose

$$\lim_{(x,y)\to(a,b)} f(x,y) = L_1 \text{ along a curve } C_1$$

and

$$\lim_{(x,y)\to(a,b)} f(x,y) = L_2 \text{ along a curve } C_2.$$

If $L_1 \neq L_2$, then definitely we can say that the limit of the function $f(x, y)$ DOES NOT EXIST at (a, b).

So whatever path we take in order to make the variable point (x, y) to approach the point (a, b), we must obtain the same value of the limit of the function. Then only we can say that the limit of $f(x, y)$ EXISTS at the point (a, b). In other words, the **independence of path** must be maintained. Figure 6.2.3 illustrates this.

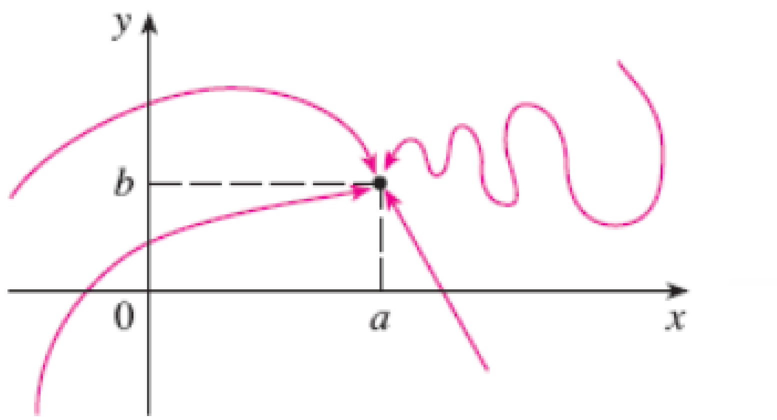


Figure 6.2.3: Independence of path

Example 6.2.2. Recall the function $g(x, y)$ from Example 6.2.1. We had

$$g(x, y) = \frac{x^2 - y^2}{x^2 + y^2}.$$

In Example 6.2.1, we had intutively concluded that the limit of $g(x, y)$ does not exist as (x, y) approaches the origin. Let us now see a formal proof of this fact. Observe that

$$\lim_{(x,y)\to(0,0) \text{ along } y-\text{axis}} \left(\frac{x^2 - y^2}{x^2 + y^2}\right) = \lim_{y\to 0} \frac{-y^2}{y^2} = -1,$$

as we have $x = 0$ along the y-axis. And

$$\lim_{(x,y)\to(0,0) \text{ along } x-\text{axis}} \left(\frac{x^2 - y^2}{x^2 + y^2}\right) = \lim_{x\to 0} \frac{x^2}{x^2} = 1,$$

as we have $y = 0$ along the x-axis.
Since $-1 \neq 1$, we can conclude that the limit does not exist.

Another way to prove that the limit does not exist is the following: Let (x, y) approach the origin along the straight line $y = mx$. Then we have:

$$\lim_{(x,y)\to(0,0) \text{ along } y=mx} \left(\frac{x^2 - y^2}{x^2 + y^2}\right) = \lim_{x\to 0} \frac{x^2 - m^2x^2}{x^2 + m^2x^2} = \frac{1 - m^2}{1 + m^2}.$$

For different values of m, that is, if (x, y) approaches (a, b) through different straight lines having slope m, we will get different limits. So we can conclude that the limit does not exist. $\square$

Remark 6.2.3. It is always easy to prove that a limit does not exist by considering two or more different paths. But it is NOT EASY to prove that a limit exists by considering the path method. That's because, there are infinitely many paths via which the variable point (x, y) can approach the point (a, b). So for proving that a limit exists, we should always go by the $\epsilon - \delta$ definition (given earlier in this section).

Example 6.2.4. Let $f(x, y) = \frac{3x^2y}{x^2+y^2}$. Find the limit if it exists at the point $(0, 0)$.
Now if we consider the limit of f along the x-axis or the y-axis or along the $y = mx$ line, it is easy to see that all these 3 approaches will give 0 as the value of the limit. But there are infinitely many directions in which (x, y) can approach the origin. Since these 3 approaches were giving us the same value 0 as the limit, therefore we may assume (for the time being) that $L = 0$ and verify it via the $\epsilon - \delta$ definition. That is, we have to verify/prove the following:

Given any $\epsilon > 0$, there exists a $\delta > 0$ such that

$$|\frac{3x^2y}{x^2 + y^2} - 0| < \epsilon \text{ whenever } 0 < \sqrt{(x - 0)^2 + (y - 0)^2} < \delta.$$

Observe now that

$$|\frac{3x^2y}{x^2 + y^2} - 0| = \frac{3x^2|y|}{x^2 + y^2}.$$

Since $x^2 \leq x^2 + y^2$, we have $\frac{x^2}{x^2+y^2} \leq 1$. This implies that

$$|\frac{3x^2y}{x^2 + y^2} - 0| = \frac{3x^2|y|}{x^2 + y^2} \leq 3|y|.$$

But $3|y| = 3\sqrt{y^2} < \epsilon$ whenever $0 < \sqrt{x^2+y^2} < \frac{\epsilon}{3}$. This is because $\sqrt{y^2} \leq \sqrt{x^2+y^2}$. So, if we choose $\delta = \frac{\epsilon}{3}$, then we can see that we can readily prove/verify the required statement. Hence we conclude that

$$\lim_{(x,y)\to(0,0)} f(x,y) \text{ exists and equals } 0.$$

$\square$

—————-END—————